Networks and Complexity

Exercise Sheet 15: Social Distancing

This is an exercise sheet from the forthcoming book Networks and Complexity.
Find more exercises and solutions at https://github.com/NC-Book/NCB

Ex 15.1: Trace and determinant [Lvl. 1]

Compute the trace and determinant of the following matrices:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Ex 15.2: Network SIS model [Lvl. 2]

Consider the network SIS model given by

$$[\dot{I}] = pz(1 - [I])[I] - r[I]$$

Compute the steady states and use the result to find the critical value of p from which on the disease can spread. Then, find the critical value of p in another way, by finding the point where the stability of the disease-free state changes. Do the results agree?

Ex 15.3: Individuality [Lvl. 2]

Consider a network of agents which can be in either of two states A, or B. No agent wants to be in the same state as its neighbors. Links in the network are checked at rate r. When a link is checked, the agents connected by the link compare their state. If they are in the same state, one of them chosen randomly switches to the respectively other state. For example, if an AA-link is checked one of the agents would switch to state B. Write a mean field model, find its steady state and check the stability.

Ex 15.4: aSIS moment expansion [Lvl. 2]

In the aSIS model infected nodes recover at the rate r, the infection spreads across SI links at rate p and susceptible nodes rewire their SI links from infected neighbors to randomly chosen susceptible partners at rate w, derive a differential equation for the density of SS-links, $[SS]$, as a function of the density of SI-links, $[SI]$, and SSI-triplet chains, $[SSI]$ (i.e. derive $[\dot{SS}] = r[SI] - p[SSI] + w[SI]$.)

Ex 15.5: Competing epidemics [Lvl. 3]

Consider two competing SIS diseases I and J, which spread at different rates $p_1 \neq p_2$ but from which people recover at the same rate r. Assume that being infected with one of the diseases gives the infected person immunity to the other disease. Use a mean field approximation to show that the system cannot be in a stationary state where both diseases persist.

Ex 15.6: aSIS invasion threshold [Lvl. 3]

The aSIS model is described by the following differential equation system:

$$\begin{aligned} [\dot{I}] &= p[SI] - r[I] \\ [\dot{SS}] &= r[SI] - p[SSI] + w[SI] \\ [\dot{II}] &= p[SI] + 2p[ISI] - 2r[II] \end{aligned}$$

along with the two conservation laws $[S]+[I]=1$ and $[SS]+[II]+[SI]=z/2$ and the closure approximations

$$[SSI]=\frac{2[SS][SI]}{[S]} \qquad [ISI]=\frac{[SI]^2}{2[S]}.$$

a) Compute the Jacobian matrix $\mathbf{J}$ that describes these differential equations.
b) Find a condition for the 'invasion threshold' from which on the disease can invade a disease-free system. Hint: Use $|\mathbf{J}|=0$.
c) Find the critical value of p from which on invasion of the disease is possible. As a test consider the case $w=0$ and compare it to the condition for the non-adaptive SIS model from Ex. 15.2.

Ex 15.7: aSIS steady states [Lvl. 3]

Compute the steady states of the aSIS model, studied already in Ex. 15.6. This will require good 'navigational skills'. If you get lost check the solution.

Ex 15.8: adaptive Voter Model (aVM) [Lvl. 3]

Consider a network with N nodes and K links, where nodes represent agents and links represent social contacts. Each agent is in either of two states A and B, say favoring one of two political parties. AB-Links, connecting agents of different opinion, get updated at rate r. In an update with probability p the link is cut and one of the two agents (chosen randomly) creates a new link to an agent in the same opinion state (so the new link is an AA or BB link). Otherwise (so with probability, $\bar{p}=1-p$) one agent chosen at random convinces the other of their opinion.

a) Derive a mean-field model for $[A]$ and $[B]$, the density of agents holding opinion A and B.
b) Derive differential equations for $[AA]$ and $[BB]$ and close using a pair approximation. (Don't use the conservation law yet.)
c) Consider the case $[A]=[B]=1/2$ and find the two steady states of the system (Hint: You will need to use a conservation law, but delay this as long as possible to preserve the symmetry)
d) In simulations of this model, a fragmentation transition is observed. On the one side of this transition there is ongoing opinion dynamics which can observed over a very long time. On the other side the system rapidly splits into two separate components such that no agent of opinion A has contact to an agent with opinion B. Estimate the value of p at which this transition occurs. (Bonus: Your estimate from the moment expansion won't be very precise in this case, formulate an hypothesis why this might be the case.)

Ex 15.9: aSIS invasion threshold done differently [Lvl. 4]

Consider a typical newly infected node in the aSIS model in the scenario where the disease is still very rare. Find a way to compute the secondary infections R_0 that this node causes before it recovers. Does this give you the same threshold for disease invasion as above? (Because other susceptible nodes will rewire their links away from your focal node you may need to integrate in time. As the threshold will happen at very small infection rates you can use that p is small)

Ex 15.10: Heterogeneous Expansion [Lvl. 4]

Consider a network with N nodes and mean degree z. Existing links in this network are broken at a rate 1 per link. At rate r per node a node establishes a link to a randomly chosen partner.

a) The dynamics of the degree distribution are captured by the differential equation

$$\dot{p_k}=-\underbrace{2rp_k}_{1}-\underbrace{kp_k}_{2}+\underbrace{2rp_{k-1}}_{3}+\underbrace{(k+1)p_{k+1}}_{4} \tag{142}$$

Explain the meaning of the terms 1-4.

b) Define $G(x)$ as the generating function of the degree distribution. And derive a differential equation that captures the dynamics of G.

c) Consider the steady state of your equation and integrate to obtain $G(x)$. (Use the normalization conditions to fix the constant of integration.)

d) We have now found the generating function that the network approaches in the long run. What does it tell you about the network?

Ex 15.11: SIS ISI ODE TLA [Lvl. 4]

For the aSIS model from above. Derive a differential equation for $[ISI]$ the density of ISI-chains.

Ex 15.12: Higher closures [Lvl. 5]

It is easy to find different plausible closure approximations for motifs such as $[{}^{B}A^{C}_{D}]$, but which one is the right one?